

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (CL) Examination

May 2018

CL771 Bridge Engineering

Date: 5.05.2018, Saturday

Time: 1.30pm to 04.30p.m

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator IRC 6 and IRC 21 is allowed

SECTION-I

Q.1. Answer the following questions

- a What are the loads, forces and its effect to be considered while designing the bridges? [03]
- b Classify the IRC loadings and discuss each in detail. [03]
- c Calculate the impact factor fraction for Steel bridges of effective span 6 m subjected to IRC class B loading. [01]

Q.2. Attempt Any One [14]

- a. Calculate the design bending moment and shear force for following data.
Carriageway – two-lane 7.5m
Clear span = 6.5m
The width of the footpath 800mm on either side
Wearing coat = 90mm
Loading IRC Class 70 R(Tracked)
Width of bearing = 0.4m
Material=M30 concrete and Fe 415 steel
- b. Calculate the design bending moment and shear force for above example (Q-2a.) if Loading is IRC Class A.

Q.3. Attempt Any Two [14]

- a. Design Shear connectors for Concrete steel composite bridge with slab thickness 350mm. Symmetric steel I section is used as girder having Flange width 500mm and thickness 25mm, Web depth is 1m and thickness is 10mm, center to center girder spacing is 3.0m. The Shear force acting on composite section is 300kN. The grade of concrete is M40. Modular ratio is 13,Refer Note 2.
- b. Calculate Courbon's reaction factor and the maximum bending moment in case of T-beam bridge having the following details. The load is adjusted for maximum eccentricity as shown in figure 1.
Roadway: 2 lanes

Note 1: Important Equations.

$$\frac{x}{d} = 0.4791 \quad \lambda = 0.8$$

$$M_{ucw} = 0.45 \times f_{ck} \times \lambda \times \left(\frac{x_u}{d} \right) \times \left[1 - 0.5 \times \lambda \times \left(\frac{x_u}{d} \right) \right] \times b_w \times d^2$$

$$M_{ucf} = 0.45 \times f_{ck} \times (b_f - b_w) \times D_f \times \left(d - \frac{D_f}{2} \right)$$

Note 2: Maximum permissible pitch for the shear connector is least of

1. Three times thickness of the slab
2. 4 times the height of the stud
3. 600mm